

SCI-MAP

Ordinary Mapmaking and Observation Coaddition

Engineering Conformance Specification

Scientific contract version: v0.2

Document revision: r0.1

Scientific status: Scope-approved successor candidate; owner scientific disposition and freeze pending

Conformance status: Unassessed procedure

Canonical science source: SCI-MAP v0.2/r0.1 shared authority

Target implementation: Not selected in this document

No independent science. Every normative scientific clause and every prediction printed here is rendered from SCI-MAP-v0.2_SHARED_AUTHORITY_r0.1.tex. This document adds only evidence-recording, execution, traceability, and pass/fail procedure. If an engineering observation appears to require changing science, record a deviation and return it to scientific-owner review; do not edit the expected result in this view.

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1 Purpose and conformance target

This specification tells a reviewer how to decide whether one exact candidate conforms to SCI-MAP v0.2. It does not select a candidate, inspect an implementation, execute validation, or assert a result. Before work begins, the reviewer freezes:

- repository and immutable candidate identity;
- build profile, compiler/runtime identities, configuration, and enabled parallel modes;
- exact input fixtures and independent expected-value generators;
- numerical comparison policy and simultaneous statistical error policy;
- required product list, representation paths, log policy, and completion criteria; and
- the shared-authority digest, contract version, this document revision, owner-decision ledger revision, exact byte-identical PTC-to-MAP boundary digest, exact original-footprint-coordinate boundary, source-bound MAP profile versions, and coadd Registry record.

Evidence from a different candidate, configuration, or science-authority digest is not silently pooled. Each rerun keeps its own identity and reason. Historical operational acceptance does not substitute for conformance evidence.

2 Evidence record

Each requirement record contains at least:

Field	Required content
Requirement	Stable SCI-MAP-REQ-NNN and shared-authority digest.
Candidate	Exact source/build identity and environment profile.
Observable	Public product, typed state, API boundary, metadata, diagnostic, or failure transition under test.
Fixture	Immutable input identity and independent expected-value calculation.
Comparison	Exact predicate or preregistered numerical/statistical bound.
Result	Pass, fail, blocked by named owner decision/dependency, or not applicable with owner-approved rationale. Skipped required evidence is not a pass.
Artifacts	Machine-readable outputs, logs, traces, plots/tables where needed, and digest-addressed provenance.
Reviewer	Independent reviewer identity, date, and deviation disposition.

Failures are retained as evidence. A tolerance, fixture, expected result, or applicability decision is not changed after viewing a failure without opening a new preregistered record.

3 Claim-layer gates

Passing one layer grants no stronger layer automatically.

Layer	Minimum engineering record	Prohibited inference
Algebraic contract	Exact owner-dispositioned shared authority and internally consistent independent expected values.	No claim that a candidate implements it.
Engineering conformance	Every applicable requirement passes on one exact candidate; no required skip, unresolved deviation, or false completion.	No claim that persisted WCS/response is scientifically faithful in real data.
Representation/response fidelity	Round-trip and injection evidence covers WCS, units, response classes, boundaries, and covariance representations with preregistered bounds.	No observational photometric, astrometric, or noise performance claim.
Observational performance	Independent references and repeatability establish the named amplitude, shape, position, covariance, or false-rate metric.	No production-readiness claim.
Production readiness	Operational profile, complete products, failure/log audit, dependency closure, release controls, and owner approval all pass.	No authority beyond the exact released profile and contract version.

All nine stable MAP scope decisions are decided for this candidate. A claim may still be blocked by an unapproved source binding, an unavailable producer, or missing conformance evidence; record that state without weakening the requirement or substituting a fallback.

4 Execution order and atomicity protocol

Run evidence in the following order so later evidence cannot mask an earlier contract failure:

1. **Stage A—structural.** Parse and freeze exact identity and parent, generation, source/profile binding, producer availability, and required decision realization for frozen PTC r0.5, the exact PTC-to-MAP boundary, and the AST same- n signal coordinate, all boundaries/profiles, units, lifecycle, WCS, MAP policy, and required-product plan. When the uniform route is selected, require the exact uniform Registry, family, and handoff identities declared in the shared authority; selection alone supplies no coefficient and creates no default. Reject direct CAL, PTC-disabled, missing-product, inferred no-op, and cross-generation routes before payload retrieval.
2. **Stage B—safe numerical classification.** Retrieve only structurally admitted transformed-signal and coefficient payloads. Classify z_i as finite, non-finite, or unrepresentable and γ_i as finite strictly positive, exact zero, negative finite, non-finite, or unrepresentable. Verify coefficient availability itself imposed no finiteness requirement. Separately classify requested coverage cut as ordinary only on $[0, 1]$, expert override only when finite and greater than one, and invalid when missing, non-finite, or negative; invalid input fails before mutation.
3. **Stage C—coordinate and product.** Evaluate exact AST coordinate validity, one-hot placement/boundary, and every MAP-local required companion. For every VAL-evaluated gate retain request, applicability, eligibility, and realization axes, source binding, immutable reason, cause, and failure scope. Only requested/applicable/eligible/realized passes. Verify separately that the PTC coefficient/QC pass uses its exact PTC-owned profile and neither replaces nor rescues SCI-MAP:map_upstream_admission@2.
4. **Stages D and E—membership then accumulation.** Construct membership only after Stages A–C and perform no payload arithmetic until all required gates pass. Verify no numeric-zero mask, late masking, or aggregate sanitization substitutes for admission. Then compare exact scientific output-row identity and cardinality, plus one-pixel, one-hot half-open-boundary, prohibited-fractional, covariance, fixed-state response, procedure-response

typing, and distinct-fact observation outputs, to independent expected values; retained storage rows do not count as scientific output rows.

5. Exercise original-footprint exposure fixtures: one original influencing several descendants at different pixels; original and descendants in different pixels; missing original coordinate; boundary loss; multiple maps sharing one original; distinct observations with numerically identical original coordinates; and donor/synthesized descendants. Verify exact observation/UID/native/stable-ALIGN-slot/generation keys, layered AST ALIGN-grid parent in the target WCS, one own-coordinate placement, separate influence edges, and no precision or temporal-support promotion.
6. Verify complete immutable observation-bundle construction as a logical artifact, record its requested, effective, observation-resolved, and realized plan stages, and exercise required publication for persisted routes plus explicitly selected in-memory and coadd-only routes. In-memory retention is not a waived product.
7. Snapshot coadd-owned state; test every compatibility failure; require atomic nonmutation.
8. Evaluate all four axes of the registered observation-coadd admission aggregate profile; then exercise admitted centered-integer coaddition, exact coadd output-row identity, and companion placement. Verify the canonical target grid is chosen before construction, each input is embedded by the exact centered integer rule, and an explicit crop creates a new row-selected product with its own identity and provenance.
9. Round-trip all required representations and reconstruct each row-selected operator from provenance.
10. Exercise fixed-state, PTC full-procedure, and PTC+MAP re-resolved response families; reconstruct and disclose the complete MAP-local operator A_{MAP} and row selector Π for every valid product, while marking the whole-chain RTC-to-CAL-to-PTC-to-MAP role unavailable unless a separate approved authority establishes it; exercise an already realized PTC-grid companion applied exactly once; and honest response/covariance disclosure. For coadd covariance, require every named block before accepting the complete numerical propagation and otherwise require partial, symbolic, summarized, lineage-resolvable, or unavailable state. Verify versioned binding for later derivative products.
11. Exercise consumer claim boundaries and immutable base/unfiltered MAP parentage without assuming a closed-world registry of all future scientific uses. Verify that Pointing and OOF receive only permission to reuse the exact ordinary MAP gridding operator when their inputs conform. Their WCS/frame, model/fitting, interpretation, outputs, defaults, profiles, and qualification remain mode-owned, and no unsupplied mode profile is activated.
12. Repeat supported execution modes under the preregistered floating-point policy, while requiring exact discrete facts.
13. Only after deterministic gates pass, execute representation, external, observational, and operational evidence appropriate to the requested claim.

For every failure-injection test, capture pre-state, attempted input, emitted diagnostic, post-state, publication/cardinality state, and completion state. No partial-state comparison may omit a coadd companion merely because its payload was not the injected failure target.

4.1 Scope-discrimination cases

The following cases prevent the approved r0.1 decisions from being tested as mere prose. Each case records exact requested, effective, observation-resolved, and realized state and asserts atomic nonmutation on rejection.

Decision surface	Admitted case	Required discriminator
Uniform coefficient authority	Exact Registry/family/handoff selection and a PTC-complete same-generation payload.	Admit only through the frozen PTC-to-MAP boundary; no default, fallback, or coefficient synthesis.
Coverage cut	Finite requests at 0, within $(0, 1]$, and above 1.	Classify zero and ordinary values separately from the explicit expert override; reject missing, negative, non-finite, and unrepresentable values before mutation.
Coverage stages	Requested, effective, observation-resolved, and realized values differ in the fixture.	Preserve all four stages. With finite-positive Q_p , evaluate the separate inclusive predicates $Q_p \geq Q_{\star}c_{\text{real}}/10$ and $Q_p \geq Q_{\star}c_{\text{real}}$.

Bundle completeness	Required companions retained in memory, persisted, and consumed by a coadd-only route.	The logical bundle and plan remain complete; publication is required only for routes whose plan requires persistence.
Target grid	Inputs require centered integer embeddings into a preselected canonical grid.	Refuse post-construction grid choice or any resampling hidden as coaddition.
Crop	Explicit row selection after construction.	Emit a new identified product with selector/provenance; do not mutate or relabel the base grid.
Response	Valid fixed-state and procedure-response products.	Disclose exact MAP-local A_{MAP} , Π separately from any upstream or whole-chain claim.
Pointing and OOF	Each requests exact ordinary MAP operator reuse.	Verify only the shared operator boundary; require separate mode-owned WCS/frame, model/fitting, interpretation, outputs, defaults, profile, and qualification, and leave an unsupplied mode profile unavailable.
Application status	Contract sources compile without an identified producer candidate.	Record contract completeness only; make no application availability or conformance claim.

5 Canonical requirement conformance clauses

The following clauses are generated from the shared authority. The first paragraph is the canonical science requirement; the second is its minimum conformance route. The result field intentionally remains unassessed.

SCI-MAP-REQ-001 – Versioned authority

Canonical clause: The scientific contract version shall be v0.2. The rendered document revision shall be recorded separately as r0.1; changing layout or exposition alone shall not be represented as a new scientific contract version.

Minimum conformance evidence: Inspect both artifacts and their machine-readable provenance for distinct contract-version and document-revision fields.

Result: UNASSESSED

SCI-MAP-REQ-002 – Capability and realized PTC input

Canonical clause: The ordinary input shall be the realized SCI-PTC v0.1/r0.5 transformed calibrated- x product $z_i = Z_i^{\text{PTC}}$, a nonpolarimetric total-intensity-equivalent detector-time quantity in the inherited top-of-atmosphere, point-source-equivalent mJy/beam convention with its exact originating fixed-nominal-beam identity. Neither that unit nor a STOKES token establishes Stokes I. MAP has no CAL-to-MAP fallback, no inferred no-op PTC product, and no product when the required positive-rank PTC product is absent or PTC is disabled.

Minimum conformance evidence: Exercise exact realized PTC input, attempted CAL input, PTC-disabled and no-product routes, incompatible PTC generation, missing nominal-beam identity, and attempted Stokes-I promotion.

Result: UNASSESSED

SCI-MAP-REQ-003 – Exact occurrence identity

Canonical clause: Every candidate occurrence i shall bind observation, detector occurrence and UID, stable RTC output sample n , exact PTC product/application generation, segment, and array/network/group as applicable. Sample time is an associated attribute. Input-column, row, and container positions are locators only and shall not establish identity or equality.

Minimum conformance evidence: Permute times, containers, rows, and columns while preserving or breaking the exact parent chain; products shall follow exact occurrence identity and reject inferred joins.

Result: UNASSESSED

SCI-MAP-REQ-004 – Producer facts and MAP admission

Canonical clause: SCI-MAP shall consume PTC-owned transformed-signal availability, output-retention disposition, coefficient family/value/typed availability, and coefficient/QC proposition without reinterpretation. The PTC coefficient/QC proposition passes only when the coefficient is requested, applicable, eligible, and realized under the exact PTC-owned coefficient/QC profile for the declared MAP-facing family. It does not grant MAP use. SCI-MAP shall separately apply the source-bound occurrence profile SCI-MAP:map_upstream_admission@2; the manifest-bound SCI-VAL Registry revision binds and VAL Core evaluates that MAP-authored policy but does not author it. The immutable r0.5 profile SCI-MAP:map_upstream_admission@1 is historical, is not a compatibility alias, and shall not replace SCI-MAP:map_upstream_admission@2. Every fact, decision, cause, scope, and immutable reason remains separately typed.

Minimum conformance evidence: Toggle each producer fact and each of the four named fields independently; verify exact PTC coefficient/QC projection, separate MAP admission, preserved causes, source/profile-version rejection, and no rescue or universal producer-flag action.

Result: UNASSESSED

SCI-MAP-REQ-005 – Coordinate and projection authority

Canonical clause: MAP shall consume SCI-AST:rtc_output_grid_coordinates@1 for the same stable RTC sample n and complete parent chain as z_i ; it shall never join by time, row, cardinality, or coordinate equality. MAP owns G_{pi} through SCI-MAP:one_hot_containing_pixel@1: the unique finite in-grid pixel whose cell contains the coordinate under half-open bounds contributes $G_{pi} = 1$, every other pixel has $G_{pi} = 0$, and an outer-upper-boundary or out-of-grid coordinate contributes nowhere. Fractional projection is deferred. AST may materialize the relation only when the artifact binds the AST parent and exact MAP plan.

Minimum conformance evidence: Test exact-parent joins, centers, lower and upper cell boundaries, outer extent, invalid coordinates, sum-over-pixel normalization on its in-grid domain, and prohibited wrap/clamp/fractional placement.

Result: UNASSESSED

SCI-MAP-REQ-006 – Ordinary coefficient status

Canonical clause: The ordinary route requires one explicitly selected PTC-owned MAP-facing analysis/gridding coefficient family satisfying SCI-PTC-REQ-052-055 and SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4. The admitted Registry

SCI-PTC:analysis_gridding_coefficients@draft-0.1 contains SCI-PTC:uniform_constant@draft-0.1, whose exact supported value is dimensionless $\gamma_i = \omega_i = 1$, and grants MAP permission through SCI-PTC:uniform_coefficient_handoff@draft-0.1. Registration and permission do not select a family or realize a handoff. MAP shall use this family only after explicit selection, complete atomic PTC publication q , exact target/payload/occurrence association, and a requested, applicable, eligible, realized profile evaluation under the admitted source and compatible binding generation. MAP does not reproduce, select, or infer the PTC-owned formula. An absent selection without an authorized mode default, unregistered family, incomplete PTC publication, missing or mismatched payload, unavailable coefficient/QC decision, or incompatible generation makes the affected route unavailable; there is no unity or alternate-family fallback. Structural availability remains distinct from finiteness. MAP alone classifies an authorized payload as finite positive, exact zero, negative finite, non-finite, or unrepresentable and applies only a matching finite strictly-positive value. The uniform family supplies no precision, sensitivity, loading, scatter, response, covariance, exposure, validity, optimality, or default meaning.

Minimum conformance evidence: Exercise the registered-but-unselected, selected-but-unpublished, missing/mismatched association, nonpassing four-axis evaluation, incompatible generation, and exact admitted uniform cases; verify producer QC, MAP admission, numerical classification, contribution, and every identity, support, lifecycle, uncertainty, and prohibited meaning separately.

Result: UNASSESSED

SCI-MAP-REQ-007 – Exposure lineage and accounting

Canonical clause: ALIGN-owned e_{acq} and valid-original e_{vo} , in seconds, shall remain immutable on stable original occurrences through RTC/CAL/PTC. The products are normatively **upstream_eligible_original_footprint_exposure** and **retained_original_footprint_exposure**. The shorter prose labels “upstream-eligible exposure” and “retained exposure” are explicit aliases for those products and no others. MAP shall deduplicate by exact observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation; place each positive- e_{vo} original exactly once through SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1, its layered AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS, and SCI-MAP:one_hot_containing_pixel@1; and never use a descendant RTC-output coordinate. Geometry/pointing realization, frame, coordinate validity, boundary state, parent, and version are required. Upstream-eligible and retained exposure differ only by ancestor-population restriction. Causal influence, representative RTC occurrence, and original-footprint exposure remain separate. These exposure products are not complete temporal support, effective map integration time, precision, or normalized-map influence. Missing original-coordinate authority makes the affected exposure role unavailable. Invalid-original has $e_{vo} = 0$; synthesis, replacement, donor reuse, overlapping filters, and decimation create no exposure. Boundary loss contributes nowhere. Coadd exposure unions atomically admitted observation-scoped original identities.

Minimum conformance evidence: Use one original influencing several descendants at different pixels, an original and descendants in different pixels, missing original coordinate, boundary loss, multiple maps sharing one original, distinct observations with identical coordinates, donor and synthesized descendants, and repeated lineage; verify exact own-coordinate placement, deduplication, and separate influence support.

Result: UNASSESSED

SCI-MAP-REQ-008 – Response information state

Canonical clause: Every valid MAP product shall disclose its known MAP-local realized operator response $A_{MAP,\Pi}$, including exact input/output domains, membership, projection, coefficient, support-row, WCS, parent, lifecycle, and limitations. MAP shall separately type upstream-conditioned fixed-state linear response, PTC full-procedure finite differences, and PTC+MAP re-resolved procedure response. An available fixed-state object shall also bind source domain, PTC-sample codomain, units, parents, response basis, and upstream limitations; a full-procedure object shall additionally bind perturbation family α and a separately typed ΔS_{PTC-FP} state-change record. The PTC+MAP family reruns PTC and resolves Π_0 and Π_+ . A whole-chain RTC-to-CAL-to-PTC-to-MAP response is reserved for a separately authorized complete study and is unavailable unless so established. Absent upstream response shall be reported as limited or unavailable without concealing the known MAP-local operator, invalidating an otherwise valid base map, fabricating a whole-chain identity, or prohibiting later analysis. A later response estimate or corrected map is a new immutable versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Exercise MAP-local-known/upstream-unavailable, fixed-state, PTC full-procedure, PTC+MAP re-resolved, companion-starting-at-MAP, partial, and attempted whole-chain cases; verify exact typing, application exactly once, complete signal membership, honest limitations, and no fabricated identity.

Result: UNASSESSED

SCI-MAP-REQ-009 – Immutable plan and lifecycle

Canonical clause: The accepted request shall be immutable and shall resolve one way through requested, effective, observation-resolved, and realized states. Disabled expert values remain inactive; one observation’s resolved state shall not leak into another.

Minimum conformance evidence: Compare serialized states across two deliberately different observations and verify

one-way provenance and complete replacement.

Result: UNASSESSED

SCI-MAP-REQ-010 – Factored contribution predicate

Canonical clause: For each pixel, membership in C_p shall be the conjunction of ten exact Boolean pass predicates derived from separately retained typed records: transformed-signal availability; PTC output retention; coefficient availability; PTC coefficient/QC proposition pass $_{\gamma, QC}$; MAP upstream admission; AST coordinate validity; finite z_i ; finite strictly positive γ_i ; admitted projection/boundary relation; and MAP-local contribution plus required-companion predicates. Every VAL-evaluated gate retains named request, applicability, eligibility, and realization fields; only request=**requested**, applicability=**applicable**, eligibility=**eligible**, and realization=**realized** passes. Every other field combination is nonpassing with its exact cause, failure scope, and immutable reason retained. A structural source- or profile-binding failure is recorded only when that structural condition actually occurs; an ordinary nonpassing field value is not thereby a structural failure. Evaluation follows the ordered DAG: A structural identity/parent, generation, source/profile, producer availability, and decision-realization checks; B retrieval only of structurally admitted payloads and MAP classification of z_i and γ_i ; C AST validity, one-hot boundary/placement, and MAP-local required companions; D membership construction; E accumulation. All gates pass before payload arithmetic. Reading an authorized payload to classify it is not arithmetic, and excluded non-finite values never contaminate accumulation. T/F/U/C or four-field records shall never be conjoined directly; no numeric-zero mask represents exclusion.

Minimum conformance evidence: Vary one gate, each of the four named fields, source-binding failure, and knowledge state at a time; compare the typed record, Boolean projection, exact DAG stage, membership, cause/scope, and immutable reason; place NaN/Inf behind structurally excluded and admitted cases to distinguish safe classification from arithmetic contamination.

Result: UNASSESSED

SCI-MAP-REQ-011 – Ordinary accumulators

Canonical clause: On the fixed contribution set, SCI-MAP shall accumulate $N_p = \sum_i a_{pi} z_i$, $Q_p = \sum_i a_{pi}$, and, when available for the identical membership, $K_p = \sum_i a_{pi} k_i$, where $a_{pi} = G_{pi} \gamma_i > 0$.

Minimum conformance evidence: Use one-pixel and one-hot hand calculations and compare accumulators before normalization; fractional projection is not authorized in v0.2.

Result: UNASSESSED

SCI-MAP-REQ-012 – Normalized publication

Canonical clause: The scientific normalized output vector shall contain exactly the rows $p \in \mathcal{S}_{out}$ authorized by the applicable effective support policy. On those rows finite $Q_p > 0$ gives $\hat{m}_p = N_p/Q_p$, and an available response companion gives $\hat{k}_p = K_p/Q_p$. No unsupported row, zero-filled value, or sentinel is a normalized scientific output or measured zero.

Minimum conformance evidence: Exercise positive, exact-zero, near-zero, overflowed, and non-finite aggregates; compare exact row identity and inspect any nonauthoritative storage slots.

Result: UNASSESSED

SCI-MAP-REQ-013 – Accumulator interpretation

Canonical clause: The numerator is a pre-normalized accumulator and Q_p is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

Minimum conformance evidence: Audit schema, metadata, API types, and labels for every numerator and normalization representation.

Result: UNASSESSED

SCI-MAP-REQ-014 – Constant preservation

Canonical clause: With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

Minimum conformance evidence: Map constant fixtures with uniform and unequal coefficients; compare exact output-row membership, then confirm that stronger response claims remain separately gated.

Result: UNASSESSED

SCI-MAP-REQ-015 – Projection-specific estimand

Canonical clause: The authorized base projection is exactly `SCI-MAP:one_hot_containing_pixel@1`; it preserves unity over the finite in-grid domain and has explicit half-open boundary loss. Fractional projection, diagonal weighted least squares, and conservation across a finite boundary are outside this branch until separately authorized.

Minimum conformance evidence: Use containing-pixel and outer-boundary fixtures; attempt fractional or WLS substitution and require method rejection.

Result: UNASSESSED

SCI-MAP-REQ-016 – One exact MAP operator for linear companions

Canonical clause: For one selected immutable realized plan Π , $A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1} J_{\text{out}} G \Omega$ is the only MAP operator and its complete realized identity is the MAP-local operator-response disclosure. Its identity binds the exact admitted PTC-sample input domain, support-authorized output-row domain, coefficient family/generation, one-hot projection plan, support policy, target WCS/grid, immutable parents, and lifecycle generation. Signal, fixed-state response, conditional covariance, an available realized PTC-grid response companion, fixed noise realizations, and a PTC full-procedure finite difference with MAP state fixed shall use that same operator, membership, placement, coefficient state, and output rows. A PTC-grid companion begins at MAP and receives the operator exactly once, never $H_{\text{fixed},\Theta}$ again. Physical original-footprint exposure instead uses unique originals and each original's own AST ALIGN-grid coordinate; causal influence is not a linear exposure companion.

Minimum conformance evidence: Trace all named uses coefficient by coefficient; compare operator domain and parent identities, verify complete MAP-local operator disclosure, prove exactly-once companion application, and separately trace original-footprint placement.

Result: UNASSESSED

SCI-MAP-REQ-017 – Realized response claim

Canonical clause: A stored kernel may be called a realized fixed-state response to template τ only when its exact sample parent satisfies $k = H_{\text{fixed},\Theta} \tau$ and the support-restricted map satisfies $\hat{k}_{\text{out}} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta} \tau = R_{\text{fixed},\Theta} \tau$ on exactly the signal rows. A PTC full-procedure finite difference $\Delta z_{\text{PTC-FP}}$ plus its separate state-change record, and a PTC+MAP re-resolved difference shall retain their distinct family/domain identities and shall not be represented by that fixed matrix. Otherwise the response is a tracer, limited, or unavailable.

Minimum conformance evidence: Inject a declared fixed-state template, PTC full-procedure perturbations, a PTC+MAP re-resolved perturbation, and a mismatched tracer; verify family identity, exact rows, state differences, and no hidden membership.

Result: UNASSESSED

SCI-MAP-REQ-018 – Absolute-offset mode

Canonical clause: SCI-MAP v0.2 shall apply no signal-centering operation: $L = I$. Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

Minimum conformance evidence: Use a constant-offset fixture and inspect both the mapped value and recorded upstream response/null-mode state.

Result: UNASSESSED

SCI-MAP-REQ-019 – Conditional covariance

Canonical clause: When a declared PTC-domain covariance N_{PTC} is available for the exact fixed membership and state Θ , MAP shall propagate it as $C_{m,\text{out}} = A_{\text{MAP},\Pi} N_{\text{PTC}} A_{\text{MAP},\Pi}^T$, retaining consequential off-diagonal terms on the exact operator domains. The equation is complete only on the declared available domain. Missing, limited, symbolic, summarized, lineage-resolvable, or unavailable terms remain typed and are never zero-filled or interpreted as independence; absence of a complete model does not invalidate the signal map.

Minimum conformance evidence: Compare a small exact propagation and honest partial, symbolic, summarized, lineage-resolvable, and unavailable states on identical operator and row membership.

Result: UNASSESSED

SCI-MAP-REQ-020 – When normalization is precision

Canonical clause: Normalization may be called marginal inverse variance only when the exact selected γ family is calibrated true inverse variance, selected errors are independent on the asserted domain, and $\sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0$. One-hot geometry alone does not prove the coefficient premise.

Minimum conformance evidence: Test one-hot geometry with both proven inverse-variance and nonprecision coefficient families.

Result: UNASSESSED

SCI-MAP-REQ-021 – Formal diagonal weight

Canonical clause: When its assumptions hold, $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \gamma_i$ is the formal marginal inverse variance. It remains distinct from Q_p , support-restricted precision $C_{m,\text{out}}^+$, exposure, hits, and empirical statistical weight.

Minimum conformance evidence: Compute and label every object for an authorized one-hot fixture and a correlated counterexample.

Result: UNASSESSED

SCI-MAP-REQ-022 – Covariance and nuisance disclosure

Canonical clause: The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Inspect every representation state, prevent promotion of normalization, hits, or diagnostic weight to covariance, and verify derivative-product binding without rewriting the original claim.

Result: UNASSESSED

SCI-MAP-REQ-023 – Data-derived state

Canonical clause: If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

Minimum conformance evidence: Use metadata cases for fixed and data-derived state and verify that unconditional labels appear only with separately admitted evidence.

Result: UNASSESSED

SCI-MAP-REQ-024 – No significance promotion

Canonical clause: Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not sig2noise or empirical significance. Empirical calibration remains NOI-owned.

Minimum conformance evidence: Attempt each prohibited relabeling through all public representations and require rejection or an unmistakable non-significance label.

Result: UNASSESSED

SCI-MAP-REQ-025 – Distinct map facts

Canonical clause: Every base/unfiltered MAP bundle shall separately represent AST-geometric incidence, MAP-route-candidate incidence, estimator-admitted contribution, admitted-observation count, `upstream_eligible_original_footprint_exposure`, `retained_original_footprint_exposure`, numerical-normalization support, science-policy support, and MAP-local base-product scientific validity. No fact or downstream eligibility decision shall become authority for another.

Minimum conformance evidence: Construct cases in which each pair differs and verify separately typed identities and immutable ownership.

Result: UNASSESSED

SCI-MAP-REQ-026 – Incidence layers

Canonical clause: AST-geometric incidence records valid coordinate placement independent of signal availability; MAP-route-candidate incidence additionally requires the exact realized PTC route and occurrence-level upstream admission; estimator contribution additionally requires all remaining gates in $\mathcal{C}_p$. The one-hot touch convention counts the unique containing pixel only.

Minimum conformance evidence: Use coordinate-valid/signal-unavailable, route-ineligible, zero-coefficient, and admitted occurrences and compare all three populations.

Result: UNASSESSED

SCI-MAP-REQ-027 – Estimator contributions

Canonical clause: Estimator-admitted contributions shall count exactly $i \in \mathcal{C}_p$. A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or with unavailable/conflicting required state, zero/negative/non-finite coefficient, non-finite signal, or out-of-grid projection, contributes nowhere; no producer flag implies a universal action.

Minimum conformance evidence: Compare exact counts with an independently evaluated profile and factored-gate oracle while preserving cause and decision scope.

Result: UNASSESSED

SCI-MAP-REQ-028 – Admitted observation count

Canonical clause: For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

Minimum conformance evidence: Use one complete, one unsupported, and one rejected observation bundle and verify the per-pixel count.

Result: UNASSESSED

SCI-MAP-REQ-029 – Upstream-eligible original-footprint exposure

Canonical clause: MAP `upstream_eligible_original_footprint_exposure` shall union and deduplicate exact positive- e_{vo} original occurrences that are ancestors of route candidates admitted by SCI-MAP:map_upstream_admission@2. Each original is placed once by its own exact AST coordinate and the one-hot half-open rule; it is not placed at every descendant coordinate. The exposure population precedes coefficient positivity and MAP-local numerical rejection. Missing original-coordinate or lineage authority makes this exposure role unavailable.

Minimum conformance evidence: Use one original influencing several candidates at different pixels, plus an eligible zero-coefficient candidate; verify one own-coordinate exposure placement, separate influence edges, and no precision promotion.

Result: UNASSESSED

SCI-MAP-REQ-030 – Retained original-footprint exposure

Canonical clause: MAP `retained_original_footprint_exposure` shall use the same original-occurrence key and own-coordinate placement as REQ-029, restricted to originals that are ancestors of estimator-admitted members. Overlapping filters, decimation, replacement, synthesis, and multiple descendants do not duplicate seconds. Coadded exposure shall union observation-scoped original identities after atomic coadd admission; one original can occupy at most one pixel under the base one-hot family, or none after boundary loss. Neither exposure is precision.

Minimum conformance evidence: Use overlapping observation lineage, multiple descendants, mixed admission, donor reuse, and boundary loss; compare exact original-key unions, pixel placement, and coadd sums.

Result: UNASSESSED

SCI-MAP-REQ-031 – Adopted threshold selector

Canonical clause: For each declared support-policy population, sort its finite strictly positive normalization coefficients Q_p ; for count $N > 0$, select zero-based $k = \lfloor ([0.75N] + N)/2 \rfloor$ and $Q_\star = Q_{(k)}$; for empty input set $Q_\star = 0$. The exact

population and policy identities and the requested, effective, observation-resolved, and realized state/value of dimensionless $c = \text{coverage_cut}$ shall be retained. The scalar shall be finite and nonnegative. Exact zero requests no relative cut, while finite strictly-positive Q_p remains mandatory. Values $0 < c \leq 1$ are the ordinary recommended range. A value $c > 1$ is admissible only as an explicit expert request and may honestly produce empty science support. Negative, non-finite, missing, or unrepresentable values shall fail before support-row construction or required-product mutation. No universal numerical default is established.

Minimum conformance evidence: Check empty and multiple population sizes against an independent integer-index oracle; verify all four state/value stages, exact-zero behavior, ordinary-range endpoints, explicit expert admission above one and possible empty science support, and preconstruction nonmutation for negative, non-finite, missing, or unrepresentable values.

Result: UNASSESSED

SCI-MAP-REQ-032 – Separate support predicates

Canonical clause: Only after the effective support policy admits the exact finite nonnegative c under REQ-031, numerical normalization support shall require finite positive $Q_p \geq Q_{\star}c/10$, and science-policy support shall independently require finite positive $Q_p \geq Q_{\star}c$. Both comparisons are inclusive and use the exact declared population and inherited order statistic. At $c = 0$, no relative threshold excludes a row, but no zero, negative, non-finite, or absent Q_p is restored. A permitted expert $c > 1$ may leave the science-support set empty. These thresholds are an operational relative-support policy and imply no exposure, precision, significance, response, completeness, or physical optimality. Neither support state establishes final validity or precision.

Minimum conformance evidence: Exercise values below, exactly at, and above both thresholds for empty and nonempty populations; distinguish normalization from science support, verify exact-zero cannot restore nonpositive/non-finite/absent normalization, and verify expert values above one without weakening the finite-positive requirement.

Result: UNASSESSED

SCI-MAP-REQ-033 – MAP-local base-product validity

Canonical clause: One MAP-local base-product scientific-validity state shall require membership in the exact support-authorized row set, exact admitted PTC and AST parents, passing MAP contribution/support policy, finite normalized signal, and absence of an applicable MAP-product failure. Explicitly limited or unavailable response/covariance does not alone invalidate the map. Retained storage, finite payload, coefficient, exposure, incidence, normalization, or downstream-local eligibility cannot establish or rewrite MAP validity.

Minimum conformance evidence: Use truth-table fixtures separating producer state, MAP admission, response/covariance disclosure, base validity, and later use-local eligibility.

Result: UNASSESSED

SCI-MAP-REQ-034 – Classify before arithmetic

Canonical clause: Only structurally admitted payloads shall be retrieved and safely classified for finiteness/sign before membership construction. All required gates shall pass before payload arithmetic. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

Minimum conformance evidence: Place NaN and Inf payloads behind structural false and true predicates; verify no unauthorized read, exact numerical classification when authorized, no contamination, diagnostics, and failure scope.

Result: UNASSESSED

SCI-MAP-REQ-035 – Exceptional contribution states

Canonical clause: Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

Minimum conformance evidence: Run one-cause-at-a-time fixtures and inspect cause-specific states plus unchanged neighboring pixels.

Result: UNASSESSED

SCI-MAP-REQ-036 – Complete observation bundle

Canonical clause: A base/unfiltered observation map shall exist as one immutable ordered logical bundle before it can participate in coaddition. The bundle contains signal, numerator, normalization identity, MAP-local realized operator response, upstream-conditioned response class/state, uncertainty/covariance state, the distinct facts in REQ-025, exposure lineage/accounting identity, unit and exact nominal-beam parent, complete AST frame/WCS, shape/indexing, estimator/projection/profile identities, PTC removed-component/additive-reference/null-space state, lifecycle, exact PTC/AST parentage, and product identity. Physical persistence is separately controlled by the effective output plan. Limited or unavailable upstream response/covariance states are complete disclosures, not automatic signal invalidity.

Minimum conformance evidence: Remove or alter each required identity or disclosure field, distinguish absent declarations from honest unavailable states, and verify that an in-memory complete logical bundle and its faithfully persisted form have the same scientific content and identity.

Result: UNASSESSED

SCI-MAP-REQ-037 – Atomic coadd admission

Canonical clause: A complete observation bundle shall pass all compatibility and required-companion checks before any coadd-owned state changes. Rejection shall leave numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

Minimum conformance evidence: Snapshot all coadd-owned state before injecting one incompatibility at a time and require byte- or field-exact nonmutation.

Result: UNASSESSED

SCI-MAP-REQ-038 – Centered integer common grid

Canonical clause: A canonical target grid may be requested and made immutable before observation-map construction. Observation maps constructed on that exact grid, or otherwise compatible observation/coadd shapes, shall use exact centered-integer embedding: shape differences are nonnegative and even in each axis, and the corresponding integer offset maps reference pixels to the same world coordinate. A later crop is a newly identified row-selected product with immutable parent and lineage; it shall not relabel the parent or make it retrospectively compatible. Fractional shift, post-hoc pad, reprojection, interpolation, wrapping, implicit recentering, mosaicking, and a different sky grid are outside this estimator.

Minimum conformance evidence: Test preconstruction target-grid requests, every parity/sign case and WCS reference-pixel relation, exact integer embedding, a separately identified post-hoc crop, attempted parent relabeling, and each excluded grid-changing operation.

Result: UNASSESSED

SCI-MAP-REQ-039 – Ordinary coadd estimator

Canonical clause: The exact v0.2 family `SCI-MAP:uniform_observation_coadd_coefficient@1`, owned by MAP, assigns dimensionless $u_{op} = 1$ to every row of each bundle eligible and realized under the VAL-governed aggregate profile `SCI-MAP:observation_coadd_admission@1`. It is an equal-observation coefficient, not precision or inverse variance. On exact centered-integer row identities, the coadd accumulates $N_p^c = \sum_o u_{op} \hat{m}_{op}$, $Q_p^c = \sum_o u_{op}$, and publishes $\hat{s}_p = N_p^c / Q_p^c$ only on $\mathcal{S}_{out}^c$. Conforming in-memory and persisted bundles have identical admission, identities, arithmetic, and result; storage route is not a scientific weighting or eligibility fact.

Minimum conformance evidence: Evaluate all four VAL fields, then use one and two observations from both in-memory and persisted conforming routes to verify equal coefficients, exact row placement, output support, lifecycle identity, arithmetic equality, and rejection of precision labels.

Result: UNASSESSED

SCI-MAP-REQ-040 – Coadd companions and unavailable response

Canonical clause: The stacked signal vector $\hat{\mathbf{m}}_{\text{obs}}$ shall carry explicit observation-row identities; B_{out} is the exact centered-integer equal-observation averaging operator; and T_{obs} is the stacked compatible fixed-state observation-response object on those same domains. Under the exact response-independent base-coadd role, response is advisory and unavailable-compatible: if any admitted member response is unavailable or basis-incompatible, the unchanged numerical signal membership may coadd only with coadd response typed unavailable, never a hidden subset or zero companion. A response-bearing coadd role instead requires every member response and exact compatible source-domain, basis, class, units, normalization, and row identity. Exposure and count follow the same atomic bundle admission, with exposure's unique-original placement.

Minimum conformance evidence: Use complete, basis-incompatible, limited, and unavailable member responses in base and response-bearing roles; verify exact role-dependent admission, unchanged base-signal membership, unavailable response, and no hidden subset.

Result: UNASSESSED

SCI-MAP-REQ-041 – Coadd covariance and precision

Canonical clause: C_{obs} shall be the block matrix on exact stacked observation-row identities with every available within- and cross-observation block explicitly located. Only when every block required by the exact covariance-qualified role is available may MAP publish the complete numerical equation $C_{c,\text{out}} = B_{\text{out}}C_{\text{obs}}B_{\text{out}}^T$. Otherwise covariance shall be explicitly partial, symbolic, summarized, lineage-resolvable, or unavailable; missing blocks are unknown, never zero or independence. The base signal coadd may remain valid under its named unavailable-compatible role. The uniform observation coefficient is not precision.

Minimum conformance evidence: Use complete within/cross blocks and each missing-block state; compare exact propagation only for the complete role and verify honest partial/unavailable disclosure without invalidating the base signal coadd.

Result: UNASSESSED

SCI-MAP-REQ-042 – Correlated-GLS boundary

Canonical clause: A correlated minimum-variance solution using $K_p^{-1}\mathbf{1}$ may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd.

Minimum conformance evidence: Use positive- and negative-correlation covariance matrices that distinguish ordinary weighting from the GLS benchmark; require method identity to remain explicit.

Result: UNASSESSED

SCI-MAP-REQ-043 – Minimum product identity

Canonical clause: Every observation and coadd product shall identify its nonpolarimetric total-intensity-equivalent role; external observation/coadd, array/group, and occurrence domains; estimator and coefficient families; MAP-local operator and upstream-conditioned response/covariance states; signal unit and fixed-nominal-beam parent; complete AST frame/WCS; preconstruction target-grid request where applicable; grid/projection/boundary; MAP profiles; exact PTC product/application and coefficient generations; exposure convention; logical-bundle and plan-controlled persistence state; lifecycle; parentage; causes; and product identity. No STOKES token supplies quantity authority. Scientific identity shall be durable without prescribing a persistence format.

Minimum conformance evidence: Round-trip every identity through in-memory and persisted representations, detect collisions or generation mixing, and reject a bare token, storage route, or serialization convention as scientific authority.

Result: UNASSESSED

SCI-MAP-REQ-044 – Frames and units

Canonical clause: MAP shall bind the complete frame identity delivered by `SCI-AST:rtc_output_grid_coordinates@1`: exact source/target frame, reference system (including ICRS or FK5/J2000 as actually declared), equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued science axes. MAP shall not equate J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name or default. Signal/response units are `mJy/beam` with exact nominal-beam identity; covariance has its square.

Minimum conformance evidence: Inspect complete frame and unit/beam identity and require cross-frame, ambiguous-frame, cross-unit, or cross-beam coadd rejection.

Result: UNASSESSED

SCI-MAP-REQ-045 – WCS and index persistence

Canonical clause: Full-precision typed or sidcar WCS shall be the lossless admission and provenance authority. A physical FITS WCS may differ by at most 0.1 arcsec maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. FITS coordinates are one-based and memory indices zero-based.

Minimum conformance evidence: Round-trip representative centers, corners, axis signs, and index conversions; calculate maximum sky separation and exact orientation invariants.

Result: UNASSESSED

SCI-MAP-REQ-046 – Realized provenance

Canonical clause: Realized state shall record requested, effective, observation-resolved, applied, and realized identities, including all four exact `coverage_cut` stages; exact PTC/AST/boundary/profile and admitted coefficient Registry/family/handoff generations; every typed gate's named request, applicability, eligibility, and realization fields plus Boolean projection, immutable reason, cause, failure scope, and DAG stage; admitted contributions/bundles; the complete MAP-local realized operator $A_{\text{MAP},\Pi}$, accumulators, and row domains; unique-original exposure keys, stable ALIGN slots, layered AST ALIGN-grid coordinate parents, preconstruction target grid, boundary states, and separate influence edges; PTC λ , $\hat{U}_{\text{corr}}$, $U_{\text{total},\Theta}$, removed-subspace and null-space identities; fixed-state, PTC full-procedure, and PTC+MAP re-resolved response family and state-change record; covariance state; centered-integer offsets; logical-bundle and persistence-plan state; required products; and failures.

Minimum conformance evidence: Reconstruct each row-selected operator, coverage-cut transition, coefficient handoff, original-footprint exposure population, gate projection/DAG, response family, logical/persisted bundle relation, and typed unavailable state from provenance alone.

Result: UNASSESSED

SCI-MAP-REQ-047 – Failure scope and atomicity

Canonical clause: An invalid required input, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing required companion, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before live aggregate state changes.

Minimum conformance evidence: Inject each failure class at its boundary; inspect exact scope, diagnostics, nonmutation, and absence of a false completion marker.

Result: UNASSESSED

SCI-MAP-REQ-048 – Required observation products

Canonical clause: A complete logical observation bundle shall exist before coaddition. Persisted per-observation products are the normal scientific expectation and shall be published whenever the effective v0.2 output plan designates them required; every required publication shall complete or the declared operation fails without a completion claim. A plan may instead permit a fully in-memory route. A coadd-only entry may consume persisted conforming observation bundles without rerunning

upstream processing. In-memory and persisted routes shall use identical logical admission, arithmetic, product identity, and provenance. Optional nonpublication does not imply that the logical observation product did not exist, and this MAP rule does not waive PTC's upstream atomic required-publication obligations.

Minimum conformance evidence: Exercise required-publication success and failure, optional in-memory nonpublication, and coadd-only persisted input; verify identical bundle admission/arithmetic/identity, exact failure propagation, no misleading completion, and unchanged upstream PTC completion requirements.

Result: UNASSESSED

SCI-MAP-REQ-049 – Immutable base parent

Canonical clause: Downstream transformations shall preserve resolvable base/unfiltered MAP-bundle identity and MAP-local base-product validity. A later local validity/eligibility decision may differ for its named use but shall not rewrite MAP validity, alter the original claims, or sever parentage. Later response or covariance estimates are new versioned products.

Minimum conformance evidence: Create a later locally eligible product over a MAP-invalid parent and a later estimate over a MAP-valid parent; verify both states and original claims remain immutable.

Result: UNASSESSED

SCI-MAP-REQ-050 – Complete consumer boundary

Canonical clause: No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include quantity and nominal-beam identity, estimator/normalization/coefficient/profile identity, honest response class/state and uncertainty/covariance disclosure, distinct map and exposure facts, PTC additive-reference/removed-component/null-space state, exact frame/WCS, shape/indexing, lifecycle, parentage, dependency versions, and failures. Later analyses choose their own claim requirements and may attach honest versioned response/covariance products; MAP does not own their eligibility.

Minimum conformance evidence: Exercise claims with missing required facts and analyses that explicitly accept limitations; verify fail-closed claim scope without rewriting MAP validity.

Result: UNASSESSED

SCI-MAP-REQ-051 – Same fixed noise operator

Canonical clause: An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

Minimum conformance evidence: Propagate fixed deterministic vectors through signal and realization paths and compare row identities and operator coefficients, not filenames or shapes.

Result: UNASSESSED

SCI-MAP-REQ-052 – Method and claim separation

Canonical clause: Pointing and OOF may be registered consumers of the exact ordinary MAP gridding operator only when their inputs satisfy this operator contract. Reuse grants no mode-specific WCS/frame, source model, fitting, response interpretation, output, default, or qualification authority; an unsupplied mode profile remains unavailable. This authority shall not otherwise be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Independent inspection, modification, experimentation, and proposal remain unrestricted, while an altered scientific identity or official claim requires an explicit versioned successor and evidence appropriate to that claim. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

Minimum conformance evidence: Exercise conforming and nonconforming Pointing/OOF operator requests, missing mode profiles, each prohibited method/claim promotion, and an independently changed support algorithm; verify exact operator reuse boundaries, unavailable unsupplied authority, versioned identity, and independent claim-state fields.

Result: UNASSESSED

6 Prediction execution matrix

Predictions are imported from the same shared authority. Engineering records attach fixtures, bounds, artifacts, and results; they do not rewrite the prediction.

SCI-MAP-PRED-001 – Uniform constant

Canonical prediction: For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized MAP-valid output row; numerator and normalization scale together, and no unsupported scientific row is created.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-002 – Unequally weighted constant

Canonical prediction: For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-003 – One-pixel hand calculation

Canonical prediction: For an authorized row with admitted z_i and $a_i > 0$, the exact triplet is $(N, Q, \hat{m}) = (\sum a_i z_i, \sum a_i, \sum a_i z_i / \sum a_i)$; an unauthorized row has no normalized scientific value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-004 – One-hot boundary

Canonical prediction: A finite coordinate inside exactly one half-open cell has one $G_{pi} = 1$, preserves unity over pixels, and contributes only there. A coordinate on an internal upper edge belongs to the adjacent cell; a coordinate on the outer upper extent or outside the grid contributes nowhere. Fractional splitting is not authorized.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-005 – Delta input

Canonical prediction: For a declared fixed state, a basis-pixel sky perturbation produces the corresponding support-row-restricted column of $R_{\text{fixed}, \Theta} = A_{\text{MAP}, \Pi} H_{\text{fixed}, \Theta}$, including projection and boundary truncation; a PTC full-procedure or PTC+MAP re-resolved difference is separately typed and need not equal that column.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-006 – Point-source template

Canonical prediction: A declared point-source perturbation τ_b produces $R_{\text{fixed}, \Theta} \tau_b$ only for the exact fixed response state and authorized rows; PTC full-procedure and PTC+MAP re-resolved families retain their perturbation/state parameters. The unavailable whole-chain RTC-to-CAL-to-PTC-to-MAP family is not inferred. Peak, integrated, and fitted amplitudes remain distinct functionals.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-007 – Resolved and Fourier inputs

Canonical prediction: At fixed state, resolved disks, gradients, and Fourier modes produce their explicit $R_{\text{fixed}, \Theta} \tau$ on the authorized row set; a constant-input pass does not predict preservation of these modes, and procedure-response families are not inferred from the fixed matrix.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-008 – Variable coverage

Canonical prediction: Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-009 – Finite boundary

Canonical prediction: At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-010 – Coefficient states

Canonical prediction: The admitted uniform family yields exact dimensionless $\gamma_i = 1$ only for an explicit selection with complete PTC publication, exact target/payload/occurrence association, MAP permission, and a requested, applicable, eligible, realized named handoff under the exact compatible source binding. Registry presence alone realizes nothing. Missing selection or handoff remains unavailable with no fallback. Structural availability is independent of numerical class; MAP alone classifies finite positive, exact zero, negative finite, non-finite, and unrepresentable. Zero is noncontributing and the other nonpositive/non-finite classes are invalid with exact cause; uniform one is not precision.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-011 – Use-qualified exclusion and non-finite states

Canonical prediction: A candidate whose typed gate record projects to Boolean nonmembership does not contaminate an accumulator. Structural exclusion prevents payload retrieval; structurally admitted payloads are read only for safe MAP numerical classification before membership and arithmetic. Its complete named-field or other typed record, immutable reason, and cause remain preserved; no universal producer-flag action is inferred.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-012 – Threshold selector

Canonical prediction: For each predeclared positive finite coefficient population, the selected zero-based index is exactly $\lfloor ([0.75N] + N)/2 \rfloor$; empty input yields $Q_\star = 0$, but no row gains support without its own finite positive Q_p . At exact $c = 0$, both relative thresholds are zero and the finite-positive requirement still excludes zero, negative, non-finite, or absent normalization. Values $0 < c \leq 1$ use the same inclusive comparisons in the ordinary recommended range. An explicitly requested expert $c > 1$ may yield empty science support. Negative, non-finite, missing, or unrepresentable c fails before support rows or required-product mutation, with all four exact state/value stages retained.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-013 – Missing response

Canonical prediction: Every valid map discloses its known MAP-local realized operator response on the exact signal rows. Missing upstream response yields a separately typed limited or unavailable upstream-conditioned state on the complete signal membership, never a smaller hidden subset, zero, or identity. Fixed-state, PTC full-procedure, and PTC+MAP re-resolved families remain distinct; whole-chain RTC-to-CAL-to-PTC-to-MAP response remains unavailable unless separately established. The base MAP bundle remains usable for claims that do not require the absent response; any later response estimate or corrected map has a new versioned identity bound to the original.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-014 – Atomic rejection

Canonical prediction: A missing required companion or any incompatible identity, unit, response, WCS, shape, policy, or parent rejects the entire observation before every coadd accumulator and count, including apparently compatible pixels, is changed.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-015 – One-observation coadd

Canonical prediction: A coadd containing one admitted observation under the uniform dimensionless coefficient family reproduces its signal on corresponding centered-integer authorized rows. Conforming in-memory and persisted logical bundles give the same admission, identity, and arithmetic. It reproduces upstream-conditioned response only when the exact response companion is available; otherwise that response remains typed unavailable while the MAP-local operator disclosure remains known.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-016 – Uniform observation coefficients

Canonical prediction: On each authorized coadd row, two compatible observations produce their equal-observation arithmetic mean and count-like normalization under `SCI-MAP:uniform_observation_coadd_coefficient@1`. The normalization is not precision.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-017 – Missing or invalid observation

Canonical prediction: An unsupported observation row contributes at no coadd row; an incompletely admitted observation contributes nowhere and does not increment admitted-observation count. Neither case creates a zero-valued scientific substitute.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-018 – Centered integer placement

Canonical prediction: A canonical target grid requested before construction yields observation maps already bound to that grid. Other compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation. A later crop is a new row-selected product with its own identity and lineage and cannot relabel its parent as compatible.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-019 – Cross-observation covariance

Canonical prediction: For two maps with variances σ_1^2, σ_2^2 and covariance c , a complete covariance-qualified uniform coadd includes c . If a required block is missing, covariance is partial, symbolic, summarized, lineage-resolvable, or unavailable rather than a complete numerical matrix; the missing block is never zero or independence, and equal-observation normalization is not precision.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-020 – Fixed noise operator

Canonical prediction: A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient $A_{\text{MAP},\Pi} \equiv A_{\text{out}}$ and B_{out} as the signal path; recomputed state exposes a named difference.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-021 – Sequential and parallel reduction

Canonical prediction: Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-022 – No false significance

Canonical prediction: A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-023 – No validity promotion

Canonical prediction: A downstream product may have its own use-local validity or eligibility but cannot change MAP-local base-product validity or sever the base/unfiltered MAP parent.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-024 – No ordinary predicate for JINC

Canonical prediction: A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.2.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-025 – Aggregate and index failure

Canonical prediction: An exact-zero, non-finite, overflowed, or unrepresentable aggregate/index never becomes a live valid product; failure occurs at the declared scope before mutation or completion.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

7 Mechanical completeness checks

Before a conformance verdict, automated checks shall establish:

- exactly one declaration of each stable requirement and prediction ID in the shared authority, with no gaps or duplicates;
- every requirement appears in this specification and in `CROSSWALK.md`; every prediction is linked to at least one requirement;
- each rendered PDF identifies contract v0.2 and its current document revision separately;
- no unresolved LaTeX reference, citation, overfull box, clipped glyph, broken table, or unreadable page remains;
- every required artifact reopens successfully, has the expected page count, and renders every page through an independent PDF renderer; and
- the owner-decision ledger and author-draft decision record contain no silent resolution of an open scientific question.

The conformance verdict is **pass** only when every applicable requirement has passing evidence on the exact candidate and all requested stronger claim layers have their own passing evidence. Otherwise the record is fail, blocked, or incomplete. This document itself records no verdict.

8 Required handoff contents

An engineering handoff contains the candidate and authority identities; requirement result table; prediction result table; exact commands and environment; independent expected-value artifacts; build/test/log outputs; representation and observational evidence where claimed; failure-injection snapshots; skipped/blocked items; deviations and owner dispositions; and an explicit statement for each claim layer. The handoff shall not collapse “conforms algebraically,” “represents faithfully,” “performs on sky,” and “ready for production” into one status word.